

BORDER OPTICS — POORI RESEARCH SAMJHO

PDF 20 of 22 — Research Paper, Part 1

**Abstract, Introduction, Literature Review, Methodology (Sections
1 se 3.12 tak)**

Sakshi ke liye, Sakshi ki apni research par. Ab tak ki 19 PDFs mein humne har piece alag-alag seekha — geocoding, satellite extraction, statistics, triangulation, visualization. Ab hum ek kadam peeche hat kar dekhte hain: wahi saari pieces jab ek formal academic research paper ki shakal mein likhi jaati hain, to kaisi dikhti hain. Ye PDF `BO_Research_Paper.md` ke pehle hisse ko — Title, Abstract, Introduction, Literature Review, aur poori Methodology (Section 3.1 se 3.12 tak) — line ki tarah nahi balki paragraph ki tarah, ek-ek dava (claim) pakad kar samjhata hai.

Ye ek code PDF nahi hai — isliye 'har line do baar' wala rule yahan apply nahi hota. Iski jagah, har academic claim ko do baar samjhaya jaayega: (1) wo claim kehti kya hai, aur (2) wo claim humari kis pichli PDF ki mehnat se aayi hai. Paper khud bhi apni history bahut khule dil se batata hai — kahaan galti hui, kahaan pakdi gayi, kahaan fix hui — aur hum wahi kahaniyan yahan bunenge.

Shuru Karne Se Pehle — Ek Research Paper Hota Kya Hai

Socho tumne ek poori building bana di — foundation, walls, wiring, plumbing, sab kuch, ek-ek karke, mahino mein. Ab agar koi aake sirf poochta hai 'ye building kaisi hai', to tum use pura construction-site ka tour nahi dete — tum ek clean, organized presentation dete ho: yahan entrance hai, ye floor plan hai, ye safety-certificate hai. Research paper bhi wahi kaam karta hai. Tumne BORDER_OPTICS mein mahino tak code likha, bugs fix kiye, numbers nikale — wo sab 'construction' tha. Paper us construction ka final, organized presentation hai, jo ek stranger (jaise koi reviewer, ya Parliament ka koi member) bina poori code padhe bhi samajh sake.

Is paper ka poori title hai: *'Satellite Verification of India's Vibrant Villages Programme: A Compositing Window Robustness Assessment of Border Village Development.'* Ye title do hisso mein tuta hua hai, aur wo split jaan-bujh kar hai. Colon se pehle wala hissa batata hai kis cheez ka satellite se verification ho raha hai — Vibrant Villages Programme (VVP-I), Border ke gaon-development ki scheme. Colon ke baad wala hissa batata hai research ka asli star kya hai — 'Compositing Window Robustness Assessment.' Ye dusra hissa hi is poori research ki sabse bada twist hai, jo hum Section 3.5 mein dekhenge: ek hi satellite data ko do tarikon se 'compose' karne se poori nateeja palat jaata hai.

Is PDF Ka Scope

BO_Research_Paper.md ke pehle 161 lines — Title, Abstract, Section 1 (Introduction), Section 2 (Literature Review, 2.1 se 2.4), aur Section 3 (Data and Methodology, 3.1 se 3.12) — bilkul 'Section 4. Results' shuru hone se theek pehle tak. Results, Discussion, Limitations, Future Work, aur Conclusion PDF 21 mein aayenge.

Section — Abstract: Poore Paper Ka Nichod, Ek Paragraph Mein

Abstract kisi paper ka sabse mushkil hissa hota hai likhna — kyunki isme poore mahino ki mehnat ko sirf 4 paragraphs mein daalna hota hai, bina kuch bhi galat simplify kiye. BORDER_OPTICS ka abstract khaas taur par dense hai, kyunki ismein har headline number apni jagah justify karta hai. Hum ise paragraph-by-paragraph todte hain.

Paragraph 1 — Scheme Ka Introduction Aur Headline Null Result

Abstract ki pehli line hi seedha vaar karti hai: agar Bharat sarkar se poocha jaaye ki unki flagship Border-village scheme kaam kar rahi hai ya nahi, to imandaar jawab — khud Parliament ke record par — 'Nahi, isse kabhi kisi ne verify nahi kiya.' Vibrant Villages Programme (VVP-I) ne lagbhag Rs. 4,800 crore manzoor kiya hai, 5 Himalayan border states/union-territories ke 2,967 gaon ke liye, jinme se 662 gaon 2022-23 se 'priority' Phase-I mein rakhe gaye hain. Is poori scheme ka koi bhi independent impact-assessment kabhi hua hi nahi.

Ye research isi khaali jagah ko bharti hai — do satellite proxies (Sentinel-2 ka NDBI, aur VIIRS night-lights) se, 251 individually-geocoded villages par teen core states (Arunachal Pradesh, Sikkim, Uttarakhand) mein, plus Himachal Pradesh ke 7 illustrative villages (kul milakar 4 states mein 258 villages), aur inhe compare kiya jaata hai ek 721-village control group se, jo usi 14 districts se aata hai lekin VVP-I mein nahi hai.

Headline Test	Full-Year Window	Summer-Matched Window
H1 — treated-only NDBI change (Wilcoxon)	p = 1.000	p = 0.435831, n = 251
H4 — control-group DiD on NDBI	coef = +0.0067, p = 0.291	coef = -0.0028, p = 0.531
Buffer sweep (250m / 500m / 1km, summer)	—	p = 0.1214 / 0.4358 / 0.8976
Leave-one-district-out (summer NDBI)	—	0 of 14 reruns significant
Randomization inference (summer NDBI)	—	p = 0.229

Ye sab numbers ek hi tarah ki kahani sunaate hain: koi bhi test, kisi bhi window mein, VVP-I ka satellite-detectable asar nahi dikhata. Lekin summer window ka p = 0.435831 wala number khud apni ek kahani rakhta hai — jo abstract khud khule dil se batata hai.

History Woven In — 154-Village Wala Reversal

Isi summer-window test ka ek pehle wala version tha, jab Sikkim ke sabhi 31 villages ka monsoon-season satellite-imagery abhi Sentinel-2 ke archive mein poora 'backfill' hoke aaya hi nahi tha. Us adhoore, 154-village subsample (Sikkim poori tarah gayab) par yahi test chalaya gaya to result aaya p < 0.000001 — ekdum saaf, jaise ek pakka, surviving signal mil gaya ho. Lekin jab poora, complete sample (sabhi 251 villages) ready hua, to ye 'signal' gayab ho gaya, aur p = 0.435831 wala null result saamne aaya. Development Log Entries 21-22 mein iska poora account hai, jisme do extraction-script bugs bhi mile aur fix hue us complete sample tak pahunchne ke process mein. Ye is poori research ki sabse important seekh hai: kabhi bhi kisi

adhoore sample ke result ko final mat maano, jab tak sample poora na ho jaaye.

Night-Lights Ka Apna Reversal

Treated-villages-only dekha jaaye to night-lights bhi wahi kahani sunata hai — dono windows mein koi significant absolute change nahi (Wilcoxon $p = 0.050$ full-year, $p = 0.9999$ summer). Lekin jab is study ne control-group list se 23 aise villages hataye jo treated villages ke hi physical ya naam se duplicate the (Development Log Entries 23-25 — ye contamination-fix story hum PDF 11/12 mein detail se seekh chuke hain), to ek full-year lights ka gap jo thodi der ke liye is study ka ek hi surviving significant result dikh raha tha, corrected, contamination-free list par null nikla ($p = 0.453$) — aur wahi leave-one-out aur randomization-inference checks bhi fail hua jo pehle isse 'holding up' ka saboot maana gaya tha.

Do Aur Layers Ki Testing — Bootstrap Aur Placebo

Is primary null ko do aur tarikon se test kiya gaya jo pehle nahi hua tha. Pehla: ek exact wild cluster bootstrap (16,384 sign-flip combinations, jo is study ke sirf-14-district-clusters wali kamzori ko seedha address karta hai) — sabhi 4 outcome/window DiD results non-significant rehte hain ($p = 0.35$ se 0.68 tak). Doosra: ek asli 2019-2021 parallel-pre-trends placebo test, jo original single pre-period control-design kabhi support nahi kar sakta tha — 4 mein se 3 combinations clean hain, chautha (summer NDBI) borderline hai ($p = 0.074$), aur ye borderline result bhi khule dil se report kiya gaya hai, chhupaya nahi gaya.

Ek aakhri correction (Development Log Entry 37): ek district-verification rerun ne control-list ke bache-khuche data-quality gap ko band kiya, do (teen mein se) affected districts ke liye — list 732 se 721 villages tak aayi. Upar diye gaye saare numbers isi corrected list par based hain, placebo test bhi (Entry 41 mein re-extract hua, koi conclusion nahi badla) — sirf ek apwaad: full-year night-lights ka baseline-balance check ab naya imbalanced nikla hai, jo pehle nahi tha (Section 6.7, PDF 21 mein aayega).

Paragraph 2 — H3: Border-Proximity Ka Ulta-Result

Built-up change, border/LAC se distance ke saath correlate karta hai (H3) — lekin securitization theory jo predict karti hai usse ulti disha mein. Summer NDBI: $\rho = 0.291$, $p = 0.0000026$. Full-year lights: $\rho = -0.252$, $p = 0.0000548$.

Ye correlation kabhi ek fresh, un-stress-tested finding tha. Ab teen checks se guzar chuka hai: leave-one-district-out (14 mein se 14 reruns significant, same sign), randomization inference ($p < 0.001$), aur ek spatial-autocorrelation-calibrated permutation test jo 251 villages ko unki asli non-independent-observations wali haqiqat ki tarah treat karta hai — aur teeno mein zinda bachta hai (spatially corrected $p = 0.0295$ summer NDBI, $p = 0.0035$ full-year lights). Isse ye ek pakka secondary

finding ban jaata hai. Kabhi ye sirf ek khula sawaal tha.

EXAMPLE — Rho Kya Hota Hai

Rho (Spearman correlation coefficient) batata hai ki do cheezein saath-saath kitni move karti hain, bina ye maane ki relationship bilkul seedhi line jaisi hi hai. $\rho = 0.291$ ka matlab: jaise-jaise border-distance badhta hai, NDBI change bhi thoda-thoda badhta hai (positive relationship). $\rho = -0.252$ ka matlab: jaise-jaise border-distance badhta hai, lights-change ghata hai — yaani border ke paas ke villages mein zyada lights-change dikh raha hai. Dono hi securitization theory ke ulte hain, jo predict karti thi ki border ke paas VVP-I ka zyada asar dikhega, isliye border-paas ke gaon zyada development dikhayenge — yahan wahi mil raha hai, lekin sirf ek proxy ki disha se, wo bhi theory ki wajah se nahi, kisi aur wajah se ho sakta hai (jaise border ke paas pehle se hi zyada infrastructure projects chal rahe hon, VVP-I se alag).

Paragraph 3 — Triangulation Ki Khuli Tension

Ek aur check ne sirf primary null ko confirm nahi kiya — isse ek khuli, unresolved tension ki tarah report kiya gaya hai, chhupaya ya smooth nahi kiya gaya. NDBI/lights ko do aur satellite proxies ke against triangulate karte hue: Sentinel-1 SAR backscatter (jo sachmuch sensor-independent hai, kyunki radar hai, optical nahi) seemat hai — null, dono windows, dono polarizations. Lekin Dynamic World ka machine-learned built-up-probability proxy aseemat hai: control-group DiD dono windows mein significant nikalta hai ($p = 0.0159$ full-year, $p = 0.0006$ summer) — aur ise nazarandaz karna mushkil hai, kyunki Dynamic World ek independent, non-satellite ground-truth (Google Open Buildings ke building-footprint counts) ke against bhi kaafi zyada strongly validate karta hai ($\rho = 0.808$) NDBI ke $\rho = 0.399$ ke muqable mein, aur teeno confirmed construction case-studies (Kaho, Walong, Musai) ko sahi pakadta hai, jahan NDBI ulti disha mein move hua tha.

Ye poora triangulation-tension ka kissa hum PDF 18 mein line-by-line seekh chuke hain — wahi Walong/Kaho/Musai news-confirmed ground-truth positive control, wahi SAR-vs-Dynamic-World disagreement. Yahan abstract mein wahi kahani ab paper ki apni awaaz mein hai.

Doosri Tension — Jo Tension Nahi Nikli, Ek Bug Nikli

Ek aur check jo pehle-pehle ek doosri khuli tension jaisa dikha — QA60 cloud-mask ko ek SCL-band alternative se cross-check karna, jisne pehli baar summer NDBI result ko null se significant mein palat diya tha — wo external review par ek coding bug nikla (SCL mask snow pixels ko 'clear' maan kar guzar raha tha). Fix karke, corrected 251-village sample par rerun karne ke baad, dono masks seemat hain, dono null ($p = 0.436$ QA60, $p = 0.288$ SCL). Paper isse saaf-saaf ek 'resolved bug caught by review' ki tarah report karta hai — ek doosri surviving tension ki tarah nahi. Ye farq zaroori hai: ek genuine tension (SAR vs Dynamic World) khuli chhodi gayi hai, ek jo sirf bug thi

(QA60 vs SCL) usse fix karke band kar diya gaya hai — dono ko ek jaisa treat karna galat hota.

Paragraph 4 — Budget Aur Closing Framing

Budget outcome ko track nahi karta ek claim banane layak tareeke mein: teeno core states ka ab valid summer-data hai, aur Arunachal Pradesh ka lagbhag das-guna bada sanctioned budget, Uttarakhand ke muqable, ek chhote mean-NDBI-change se milta hai, aur Sikkim ka sabse chhota budget ek negative change se milta hai — descriptive-only, $n = 3$, aur correlation ka sign dono compositing windows ke beech palat jaata hai.

Abstract ka aakhri paragraph poori research ki asli kahani kehta hai: jo verdict ye study pehle report karti thi, wo aaj ka verdict nahi hai — aur wahi reversal is finding ka hi ek hissa hai. Ek baar jab extraction poora ho gaya, treated/control data ek hi din pull hua, aur design ko small-cluster-bias aur asli pre-treatment placebo ke against stress-test kiya gaya, to VVP-I ka koi satellite-detectable asar bacha nahi — ek seedha, reportable jawab us sawaal ka jo Parliament ke record ke mutabik kabhi poocha hi nahi gaya tha. Independently ground-truth-validated proxy ke against triangulate karne se is primary measurement mein ek sensitivity saamne aayi — aur ye paper us sensitivity ko explain-away karne ke bajaye seedha report karta hai, saath mein ek border-proximity pattern ke saath jo poori tarah stress-tested hoke bhi khada hai. Doosri jo sensitivity lagi thi (cloud-masking convention mein), wo review par sirf ek coding bug nikli, aur kuch nahi.

Section 1 — Introduction: Kahani Ki Shuruat

Bharat ke border-villages hamesha se policy discourse mein ek particular tarah se dekhe gaye hain — liabilities ki tarah, kyunki wo door-daraz, kam-aabaadi wale, aur ek 'contested frontier' khone ke khatre mein rehte hain. Vibrant Villages Programme (VVP-I) isi sochne ke tareeke ka jawab hai — Union Cabinet ne February 2023 mein ise manzoori di, jisse Arunachal Pradesh, Sikkim, Uttarakhand, Himachal Pradesh, aur Ladakh ke 2,967 villages security-ki-afterthought se development-ki-priority ban gaye.

Lekin is kahani mein ek cheez missing hai: sanction ke do saal se zyada guzar jaane ke baad bhi, ye tay karne ka koi independent tareeka nahi ki jo development wada kiya gaya tha wo zameen par hua bhi ya nahi. Jab Parliament ne seedha poocha ki VVP ka asar hua hai ya nahi, jawab bilkul saaf tha: 'No impact assessment has been carried out' (Lok Sabha Unstarred Question No. 508, 3 February 2026, Ministry of Home Affairs).

EXAMPLE — 'Unstarred Question' Kya Hota Hai

Parliament mein koi bhi member sarkar se sawaal pooch sakta hai — likhit form mein. Do tarah ke hote hain: 'Starred' questions, jinka jawab live, sadan mein bola jaata hai aur follow-up bhi poocha ja sakta hai; aur 'Unstarred' questions, jinka jawab sirf likhit record mein daal diya jaata hai, bina live discussion ke. Ye 'Unstarred Question No. 508' isi tarah ka ek permanent, official likhit record hai — jise koi bhi Parliament ki website se access kar sakta hai. Isliye ye is research ki sabse mazboot buniyad hai: ye kisi journalist ka claim nahi, khud sarkar ka apna, likhit, permanent record hai.

Yahi research exactly wahi kaam karti hai. Village-by-village test: kya sanction ke baad zameen par koi physical development dikh raha hai; kya village ke development ka size state ke budget se correlate karta hai; kya location border ke paas hone se zyada judi hai, ya development ki asli zaroorat se. Yahin par securitization theory ka role aata hai: agar border-infrastructure ek geopolitical sanket hai, welfare policy ka measure nahi, to results waise hi predict honge jaise zaroorat ke bajaye border-closeness ke hisaab se.

Paanch Research Questions

RQ	Sawaal
RQ1	Kya priority villages mein sanction ke baad built-up area expansion hota hai?
RQ2	Kya is expansion ka size sanctioned budget se correspond karta hai?
RQ3	Kya night-lights bhi wahi built-up-change ki tasveer confirm karte hain jo NDBI dikhata hai?
RQ4	Kya development border ke paas concentrated hai, na ki jahan zaroorat sabse zyada hai?
RQ5	Kya ye poora growth VVP-I se juda hai, ya sirf village-level ka general trend hai?

Chaar Hypotheses

Ye paanch sawaal chaar testable hypotheses mein operationalize hote hain. H1: policy-implementation ke baad ka waqt built-up-area ke measure par bada farak dalta hai. H2 (implied): ye increase budget ke proportional nahi hoga — bade-budget states mein relatively chhota increase dikhega. H3: border/LAC ke zyada paas wale villages, usi state ke doosre villages ke muqable, zyada tezi se transform hote hain, jaisa securitization theory predict karti hai. H4: jo test ek programme-effect ko general trend

se alag pehchanta hai, wo predict karta hai ki VVP-I priority villages mein change, usi districts ke andar ek non-VVP control-group ke muqable, significantly zyada hoga.

In chaaron hypotheses ko hum PDF 11 (H1), PDF 12 (H4), aur PDF 13 (H3) mein already line-by-line test kar chuke hain — is PDF ka kaam unhe wapass derive karna nahi, balki dikhana hai ki wo poore paper ke narrative mein kahan fit hote hain.

Section 2 — Literature Review: Ye Research Kis Buniyad Par Khadi Hai

2.1 — Securitization Theory Aur Border Development

Buzan, Waever, aur de Wilde (1998) ki securitization theory kehti hai ki jab koi topic 'ordinary' politics se hatkar 'security' ki agenda mein daal diya jaata hai, to aise resources aur 'extraordinary' measures unlock ho jaate hain jo normally politically possible nahi hote.

EXAMPLE — Securitization Ek Analogy Se

Socho ek city mein traffic-jam ek normal, roz ki problem hai — koi urgency nahi, budget bhi limited milta hai. Lekin agar wahi traffic-jam 'national emergency' declare kar diya jaaye, to achanak unlimited budget, fast-track approvals, aur special powers mil jaate hain — kyunki ab wo 'security' ka mamla ban gaya. Border-development bhi exactly isi tarah kaam karta hai: welfare-spending ki tarah bhi legitimate hai, strategic-link-building ki tarah bhi legitimate hai — aur policy-literature mein paisa shaayad hi kabhi in dono mein alag kiya jaata hai. Ye study VVP-I ko is theory ke ek test-case ki tarah use karti hai: kya border se geographic closeness, development ka ek behtar predictor hai un conventional indicators se, jaise pehle-se-maujood infrastructure jo kahin bhi ho sakti thi?

2.2 — VVP Ki Policy Context Aur Accountability Gap

VVP-I ke scope ka poora paper-trail Parliament ke replies mein ek hi jagah nahi milta — bikhra hua hai. Poora village-wise annexure Rajya Sabha Hansard mein hai (Question No. 2321, Ministry of Home Affairs, dated 9 August 2023). Himachal Pradesh ke 75 priority villages Lok Sabha Question No. 2104 (2023) aur Rajya Sabha Question No. 3251 (2023) mein confirm hote hain, lekin naam-level list nahi di gayi. Ladakh ke 35 sanctioned villages Lok Sabha Question No. 4360 (2025) mein officially confirm hain, lekin phir bhi bina naam ke. Aur wahi sawaal jo har koi poochta hai — Lok Sabha Unstarred Question No. 508 (3 Feb 2026) mein Shri Baijayant Panda ne poocha ki VVP ka asar assess hua hai ya nahi, aur Ministry ka jawab seedha tha: 'No impact assessment has been carried out.'

Is section ka ek paragraph khud apni ek galti aur uski fix ka account rakhta hai, dated 2026-09-18. Pehle Himachal Pradesh wali confirmation Rajya Sabha Question No. 401 (3 December 2025) ko cite karti thi — jo ek asli Ministry of Home Affairs document hai, lekin ek bilkul alag topic par: Wayanad landslide disaster assistance, VVP se koi lena-dena nahi. Ek citation-audit mein ye pakda gaya, aur sahi source, Question No. 3251 (29 March 2023) se replace kiya gaya, jisme sachmuch Himachal Pradesh ke 75 villages wala state-wise breakdown hai. Isi audit mein ek doosri chhoti galti bhi pakdi gayi: closing quote pehle 'No impact assessment has been done' likha tha — ek close paraphrase, lekin verbatim nahi. Asli jawab hai 'No impact assessment has been carried out,' aur wahi ab quote mein hai.

Ye ek chhoti si history hai, lekin ek badi seekh deti hai: ek research paper mein har citation, khaas taur par government-record wale, khud dobara check karne layak hai — sirf isliye ki koi document 'sach mein exist karta hai' (jaise Q. No. 401 karta hai), iska matlab ye nahi ki wo document usi baat ke liye sahi citation hai jiske liye use kiya ja raha hai.

2.3 — Remote Sensing Approaches

Do indices is poori research ka analytical bhaav uthate hain. NDBI (Normalized Difference Built-up Index), short-wave-infrared aur near-infrared bands ke reflectance se nikala jaata hai (Zha, Gao, aur Ni, 2003), built-up surface extent ke multi-temporal comparison ke liye ek widely-used proxy hai. VIIRS Day/Night Band se milne wala radiance ek related lekin kaafi alag cheez measure karta hai — electrification aur economic activity ki matra — aur purane DMSP-OLS ke muqable behtar dynamic range aur resolution rakhta hai (Elvidge, Baugh, Zhizhin, Hsu, aur Ghosh, 2017). Ye dono indices ek-dosre ke saath achhi tarah kaam karte hain kyunki unki kamzoriyan alag-alag hain: NDBI vegetation-phenology aur snow ke against kamzor hai, VIIRS cloud-cover aur low-radiance-par-sensor-saturation ke against. Agar dono aseemat hon, to zaroori nahi ki koi ek galat ho — lekin jahan dono seemat hon, wahan baat zyada bhaari hoti hai kisi ek ke akele kehne se.

2.4 — Compositing Window Sensitivity, Ek Jaan-Bujh Kar Design Feature

High-relief terrain mein spectral-data change-detection do wajah se mushkil hai: pehla, snow-cover aur cloud-frequency season aur elevation dono ke hisaab se badalte hain; doosra, agar compositing ka tareeka elevation-sensitive na ho, to wo asli change ko ya to bana sakta hai jahan hai nahi, ya chhupa sakta hai jahan hai. Ye study is sensitivity ko 'averageable noise' ki tarah treat nahi karti — ise seedha, result ki ek khaasiyat ki tarah test kiya gaya hai, aur — zyada zaroori — ye maan kar chala gaya hai ki ye do defensible compositing windows ke beech palat bhi sakta hai. Agar in dono mein se koi ek akela confirmation ka source hota, to wahi asli test hota, na?

Section 3.5 mein hum dekhenge ki ye theek waisa hi hua.

Section 3 — Data and Methodology

Ye paper ka sabse lamba, aur is PDF ka sabse important, hissa hai — kyunki yahi wo jagah hai jahan har number ka asli 'kitchen' dikhta hai: kahan se data aaya, kaise process hua, kya-kya galat hua raaste mein, aur kaise fix hua.

3.1 — Study Design

Core sample sirf teen states ka hai: Arunachal Pradesh, Sikkim, aur Uttarakhand — 251 villages, aur sirf yahi, kyunki sirf inhi ke liye village-wise information officially cross-check ke liye maujood hai. Himachal Pradesh ke 7 confirmed villages ko ek alag illustrative case-study ki tarah likha gaya hai — wo core sample ke andar nahi baithte. Ladakh kisi bhi village-level study mein nahi hai (Section 6.1, PDF 21 mein aayega) — wahan data mein ek gap hai, isliye ye poori tarah bahar hai.

Before-after compare karne ka tareeka: ek difference-in-differences (DiD) setup (H4), district fixed-effects ke saath — matlab har district ka starting-level hata diya jaata hai, sirf trend bachta hai. Post-period ka term har district aur village mein same rehta hai. Iske upar, ek control group hai: usi 14 districts se 721 non-VVP villages, taaki jo bhi wider-region waise hi kar raha tha, VVP se juda ho ya na ho, wo bhi net-out ho jaaye. Do aur checks '2021-vs-2025' ko aage badhate hain: pehla, ek teesra time-point, 2023, beech mein; doosra, hamesha wala 500m buffer, 250m aur 1km par bhi test hota hai, ye dekhne ke liye ki kya buffer-size bhi utna hi asar dalta hai jitna compositing-window.

3.2 — Data Sources

Variable	Source	Coverage
Village ID	State VVP-I portals; Rajya/Lok Sabha Q&A; annexures	2023-2025
Coordinates	OSM Nominatim; ISRO Bhuvan Village Geocoding API	Current
Control village ID	OSM Overpass API, district-matched	Current
NDBI	Sentinel-2 SR Harmonized	2021, 2023, 2025

Night-lights	VIIRS DNB monthly composites	2021, 2023, 2025
Border/LAC geometry	Natural Earth 10m Admin 0 Boundary Lines	Current
Budget/projects	State-wise VVP-I sanction figures (parliamentary record)	2023

3.3 — Geocoding

Nominatim ne zyaadatar geocoding ka kaam kiya. Bhuvan backup ki tarah aaya, Census data se juda hua, kyunki Bhuvan ek extra check maangta hai: wo state se filter nahi karta. Arunachal Pradesh ke 'Kharman' village ke liye ek test-search ne bilkul alag, Haryana ke ek same-naam wale village ko upar la diya — ye is poori research mein woh hi cross-state false-match bug hai jo humne pehle bhi dekha hai, geocoding ke context mein. Dono sources ke har result ko haath se check kiya gaya ki wo sahi district mein hai ya nahi, aur har village-list ko official total-numbers se pehle hi match kiya gaya, tabhi accept kiya gaya.

State	Attempted	Geocoded
Arunachal Pradesh	455	186
Sikkim	46	31
Uttarakhand	51	34
Himachal Pradesh	7	7
Total	559	258

3.4 — Satellite Change Detection

Har geocoded village ke around ek 500m buffer liya gaya. Us buffer ke andar, Google Earth Engine se mean NDBI aur VIIRS night-lights radiance nikale gaye, 2021 against 2025. NDBI Sentinel-2 Level-2A Surface Reflectance se aata hai, bands B11 (SWIR1) aur B8 (NIR), clouds ko QA60 band se mask karke. Night-lights VIIRS Day/Night Band se (NOAA/VIIRS/DNB/MONTHLY_V1/VCMSLCFG).

3.5 — Compositing Window Robustness Testing

Dono measures do baar calculate hue — ek baar full-year composite ki tarah, ek baar season-matched composite ki tarah, jo seasonal noise ko subtract karta hai (full-year wale ko snow gadbad karta hai, season-matched wale ko monsoon cloud). Yahi is

double-calculation ka poora point hai — asli change ko ek seasonal artifact se alag karna. Agar kisi village ka kisi window ke liye koi valid composite nahi bana, to use zero force nahi kiya gaya — use bas null mark kar diya gaya.

History Woven In — Ye Do Windows Kaise Aaye, Aur Ek Zaroori Nuance

Development Log ke Entries 4 aur 5 batate hain ki ye poora dual-window design kaise banta gaya. Sabse pehla result (Entry 4) sirf full-year composite (Jan-Dec) se aaya, 2021-vs-2025, aur wo pehla hi result null tha — Wilcoxon $p = 1.000$, mean NDBI change bhi thoda negative. Lekin us result ko final maanne se pehle ek cheez check karni thi: ek poore-calendar-year composite mein high-altitude Himalayan terrain mein kayi mahino ka snow-cover bhi aa jaata hai, aur agar 2021 aur 2025 ke composite mein snow-covered-vs-snow-free images ka ratio alag ho — jo ho sakta hai, kyunki ye is baat par depend karta hai ki kaunse din cloud-free nikle — to wahi akela NDBI value ko hila sakta hai, zameen par kisi asli change ke bina bhi. Isi snow-cover-confound ko rule-out karne ke liye Entry 5 mein season-matched (June-September) composite try kiya gaya.

Wahan ek naya masla khada hua: summer-only window mein Sikkim ke sabhi 31 geocoded villages, dono periods mein se kam-se-kam ek mein, zero cloud-free images dete — poora Sikkim ka NDBI-data gayab ho gaya, kyunki June-September Eastern Himalaya mein peak monsoon hota hai. Jab is 154-village subsample (Sikkim ke bina) par wahi Wilcoxon test chalaya gaya, result palat gaya: $p < 0.000001$, ek highly significant increase — full-year wale result ka bilkul ulta. Yahi wo pehla reversal tha jo baad mein complete-sample ke saath dobara palta (jaisa hum Abstract mein seekh chuke).

Ek zaroori nuance, imaandaari se: paper khud Section 3.12 mein kehta hai ki ye dual-window design 'locked in before either window's result was even known' — yaani dono windows test karne ka faisla, kisi bhi window ka result dekhne se pehle hi tay ho gaya tha. Development Log ka apna, upar-diya-gaya timeline (Entry 4 phir Entry 5) dikhata hai ki full-year window pehle chalaya gaya, aur uska result ($p = 1.000$, null) pata chal chuka tha, us se pehle ki summer window try kiya jaaye — summer window to snow-cover-confound ka jawab dhundhne ke liye add kiya gaya, full-year result dekhne ke baad hi. Ye is poori paper ki baaki har history jitni hi imaandaar disclosure ka mauka hai: paper ka asli point ye hai ki dono windows ka result report karne ka faisla — sirf jo 'kaam kare' usi ko chun-kar chhapna nahi — pehle se tay tha, aur yahi bhavna sahi hai. Lekin literal wording, 'before either window's result was even known,' Development Log ke apne sequence se bilkul match nahi karti. Ye koi badi galti nahi hai — lekin 'har claim ko primary source se dobara check karo' wala rule yahan bhi utna hi zaroori nikla, jitna kisi Python script ki kisi line ke liye.

3.6 — Border Proximity Analysis

Distances GeoPandas mein nikaale gaye, sirf un Natural Earth Admin-0 segments tak jo har village ke liye Indian territory par relevant hain. Ek baat pehle keh deni zaroori hai: India aur China ke beech Line of Actual Control (LAC) contested hai — koi ek agreed-upon line nahi hai. Isliye is study mein har distance number sirf relative aur comparative hai. Koi bhi ise border kahan hai iska official statement maan kar na padhe.

3.7 — Non-VVP Control Group Aur DiD Design

Ek treated-only comparison akela jo fix nahi kar sakta: chahe VVP-I ka koi asar ho hi na, wahi rise ya fall in villages ke shared kisi cheez se bhi aa sakta hai, treatment se kuch lena-dena nahi — koi unrelated infrastructure spending, koi aur state programme, ya bas ek bure saal ke baad plain regrowth, sab data mein waisa hi dikh sakta hai.

Isi wajah se ek control group banaya gaya, dynamically, usi 14 districts se jahan treated villages hain, OpenStreetMap Overpass API se, aur VVP-I priority list mein jo bhi tha use bahar rakhte hue. Phir usi exact steps se guzra: wahi 500m buffer, wahi NDBI/VIIRS definitions, wahi before/after windows. Results ek two-period panel mein reshape hue, ek treatment-indicator ke saath, aur is tarah test hue:

$ndbi \text{ (or lights)} \sim \text{treatment} + \text{post} + \text{treatment} \times \text{post} + \text{district fixed effects}$

Purpose: Ye regression formula batata hai ki NDBI (ya lights) ka value kin factors se predict hota hai. 'treatment' — kya ye village VVP-I mein hai ya control-group mein. 'post' — kya ye 'after' (2025) period ka reading hai ya 'before' (2021) ka. 'treatment×post' — yahi asli interest ka coefficient hai: treatment aur post dono ka interaction, jo batata hai ki treated-group mein post-period mein extra kya hua jo control-group mein nahi hua. 'district fixed effects' — har district ka apna alag baseline-level allow karta hai, taaki sirf trend compare ho, level nahi. Syntax: '~' ka matlab hai 'is predicted by' — yaani left-side ki value, right-side ke terms se explain hoti hai. '+' har naya term add karta hai model mein. '×' (interaction operator) do terms ko multiply karta hai — iska coefficient hi wo number hai jo DiD ('difference-in-differences') estimate kehlata hai.

Standard errors usi 14-district variable par cluster kiye gaye. Jo number sabse zyada maayne rakhta hai, wo hai treatment×post coefficient — treated-group kitna badla aur control-group kitna badla, usi samay-avadhi mein, jabki har district ka apna baseline-growth-trend fixed-effects se hata diya gaya hai. Iske saath-saath, ek parallel HC3 heteroskedasticity-robust version bhi report hota hai, bina fixed-effects ke, kyunki 14 clusters, cluster-robust methods ke liye usual 30-40+ recommendation se kam hai.

Ek cheez jo ye design bilkul nahi kar sakta: ek poori tarah parallel pre-trends placebo test, jaisa is project ke pehle control-group-kaam ne kahin aur, ek multi-period pre-treatment panel ke saath, chalaya hai. Yahan, control-group ki satellite-extraction mein sirf wahi single before/after pair hai jo treated-sample mein hai. Isliye level-balance-check (Mann-Whitney U) hi, best case mein, ek asli baseline-check ka ek kamzor stand-in hai.

3.8 — Multi-Year Trend Extraction

Sirf do saal, 2021 aur 2025, mausam ke liye ek asaan mauka hai ek aisa change bana dena ya chhupa dena jiska development se kuch lena-dena hi nahi. Isi wajah se ek teesra time-point add kiya gaya: 2023, dono ke beech mein aadha. Ye usi 251-village core sample ke liye, dono compositing windows mein, extract hua, aur ek value per-village per-year deta hai, na ki sirf ek aur before/after delta.

Har village ke liye teen saalon ke across ek linear trend fit kiya gaya, aur Wilcoxon signed-rank test ne check kiya ki wo slope zero se alag hai ya nahi — is tarah ki time-series ke liye standard tareeka. Ek doosra, alag tareeka se wahi cheez check karne ke liye, ek village-fixed-effects panel regression (value ~ year, clustered SE by village) bhi chalaya gaya, usi slopes par. Aur do halves — 2021-se-2023 aur 2023-se-2025 — alag-alag bhi test hue, taaki koi change jo sirf ek hi half mein concentrated ho, wo pakda jaaye.

3.9 — Buffer Radius Robustness Testing

Is poori study mein har jagah 500m ek fixed choice tha, kabhi kisi aur ke against test nahi hua — ab tak. Wahi summer-window extraction 250m aur 1km par bhi repeat ki gayi, usi core sample par, aur har radius par ek Wilcoxon signed-rank test chala.

Ek complication: 250m aur 1km extractions baad ki tareekh par pull hue the original 500m wale se, chahe teeno ne identical fixed 2021/2025 date-ranges use kiye. Ye maayne rakhta hai kyunki Sentinel-2 archive hamesha aage-aage scenes 'backfill' karta rehta hai present-day ke jitna paas, isliye teeno radii ka raw comparison isi timing-gap se gadbad ho jaata — ab wo sirf buffer-radius akela isolate nahi kar raha hota. Fix ye tha: teeno radii ko sirf un villages tak restrict karna jinke paas teeno radii par data ho, taaki sample-composition fixed rahe aur sirf buffer-radius hi badle.

3.10 — Statistical Testing, Ek Recap

Test	Kahan Use Hua	Kyun
Wilcoxon signed-rank (1945)	Before/after change (H1, RQ3, multi-year trend)	Paired, non-parametric — sample normal-distributed nahi hai

Spearman rank correlation (1904)	Budget correlation (RQ2), border-proximity (H3)	Non-linear par monotonic relationships ke against robust
DiD OLS (Section 3.7)	Control-group comparison (H4)	Cluster-robust ya heteroskedasticity-robust SE ke saath

RQ2 ka budget-correlation ab teeno core states ke paas valid data hone ke baad hi poori tarah test ho paaya (summer-window extraction poora hone ke baad, Development Log Entry 22) — aur wo hamesha descriptive-only hai, jaisa Section 3.10 khud spasht karta hai.

3.11 — Causal Structure Aur Identifying Assumptions

Ye design kya control karta hai, aur kya sirf assume karta hai? Paper ka apna causal diagram (ek Mermaid DAG — directed acyclic graph, matlab ek aisi flow-chart jisme arrows ek disha mein chalte hain aur wapas loop nahi banate) ise saaf karta hai. Section 3.7 ka DiD specification sirf ek confound net-out karta hai: ek shared regional trend jo treated aur control dono villages par barabar padta hai. Baaki sab kuch bas wahin baitha hai, ek assumption ki tarah — kabhi kisi test se guzra hi nahi.

DAG Ke Nodes Aur Arrows, Prose Mein

Diagram mein: Border-proximity (BP), Elevation/terrain (EL), aur 2021 ka Baseline-development-level (BD) — teeno mil kar decide karte hain ki koi village VVP-I priority (TS) ke liye select hua ya nahi. TS aur BD dono, saath mein ek Shared-regional-trend (REG), mil kar 2021-se-2025 ka Physical-development (DEV) banate hain. DEV hi wo cheez hai jo Measured-NDBI-change aur Measured-VIIRS-lights-change ki tarah 'dikhta' hai — lekin NDBI par do confounds bhi asar daalte hain: Snow-cover (full-year window mein) aur Monsoon-cloud (summer window mein). REG aur BD, control-group-villages (CTRL) par bhi asar dalte hain, jo apna Control-NDBI-change aur Control-lights-change dete hain. Aakhir mein, treated-NDBI aur control-NDBI mil kar DIFF (treated-minus-control gap, yaani DiD estimate) banate hain.

Is diagram ko asli design ke against rakh kar dekhein: Section 3.7 ka district-fixed-effects DiD sirf REG ko net-out karta hai — jo bhi secular trend poore 14-district-region mein pehle se chal raha tha, VVP-linked ho ya na ho. Yahi wo confound hai jise ek treated-only before/after comparison (H1) akele rule-out nahi kar sakta. Lekin ye TS se wapas BP, EL, aur BD ke through DEV tak jaane wale arrow ko chhoota tak nahi. VVP-I villages sarkar ne priority-list mein chuni, randomly nahi baanti — isliye agar border-proximity, terrain, ya 2021 mein village pehle se kitna developed tha, ye cheezein treatment se alag bhi us village ki path ko shape karti, to DiD estimate use baaki sab ke saath hi soak-up kar leta hai. Apne aap se ise alag nahi

kar sakta.

District fixed-effects ise sirf district-level tak hi fix karta hai — har district ka apna fixed baseline hata deta hai. Ye kabhi district ke andar, village-level tak nahi pahunchta. Isi wajah se Section 6.7 (PDF 21) apne baseline-balance-check ko sirf ek 'level check' kehta hai, aur kuch nahi — ye ek confirmed parallel-pre-trends test nahi hai. Ek asli placebo-test ke liye CTRL aur TS branches ko ek asli pre-treatment panel ke across compare karna padega, aur Section 7.5 (PDF 21) flag karta hai ki current single-pre-period-extraction ye de hi nahi sakta.

SNOW aur CLOUD wahi compositing-window confounds hain jinke liye Section 3.5 hi exist karta hai unhe alag karne ke liye, aur unka NDBI par hi khaas asar hai (LIGHTS par nahi, kyunki wo optical imagery se aata hi nahi) — yahi wajah hai ki is study ka har built-up-area test dono windows mein chalta hai. Sirf ek window par bharosa karna kabhi kaafi nahi tha.

SUTVA — Spillover Ka Ek Conservative-Bias Argument

Diagram jo cheez explicitly nahi dikhata, wo hai SUTVA (Stable Unit Treatment Value Assumption) — ye maan lena ki ek control-village ka outcome is baat par depend nahi karta ki uske paas kaunsa village treated hua. Ye design isse guarantee nahi kar sakta. Control-group usi 14 districts mein baithta hai jahan treated sample hai, koi alag, untouched region nahi — isliye koi road, pul, ya koi bhi shared-infrastructure jo ek treated village ke fayde ke liye bani ho, wo plausibly ek geographically paas ke control-village tak bhi pahunch sakti hai, saath hi doosri, VVP-adjacent government schemes jo usi districts mein usi period mein chal rahi hon.

Agar aisa hota hai, to ye ek threat nahi hai jo yahan ek false-positive bana de. Ulta hota hai. Koi bhi aisa spillover control-group ka apna outcome upar le jaata hai, jisse measured treated-minus-control gap ghat jaata hai, badhta nahi. Matlab ye study ka null DiD result, spillover ke under, agar kuch hai to conservative hai — ek asli effect ko understate karega, ek false effect banayega nahi. Ye spillover ko rule-out nahi karta, aur seedha test bhi nahi hua yahan — lekin iska matlab hai ki jo bhi bias ye introduce karta hai, uski disha is paper ke apne headline-finding ke khilaaf kaam karti hai, uske paksh mein nahi.

3.12 — Primary Aur Secondary Estimands

Ye study bahut saare tests chalati hai — H1 do windows mein, H3 do windows mein, H4/DiD do windows aur do specifications mein, ek teen-point trend, ek teen-radius buffer-sweep, aur ek descriptive budget-comparison. Itne saare tests ho jaane ke baad, ye saaf-saaf kehna zaroori hai ki paper ka headline conclusion kis ek par khada hai — koi bhi secondary ya exploratory result chupke se uski jagah na le le.

Estimand Level	Kya Shaamil Hai
Primary estimand	H1 — treated-only NDBI change, 2021-2025, 251-village core sample, dono windows mein
Secondary, pre-specified robustness	H4 control-group DiD; buffer-radius sweep; leave-one-out; randomization inference; Holm-Bonferroni
Secondary, independent RQs	Night-lights change (RQ3); border-proximity (H3); multi-year trend; budget correlation (RQ2)
Secondary, now stress-tested	Summer-window H3 NDBI correlation — Section 4.2 correction ke baad hi saamne aaya

Primary estimand — 251-village core sample par, 2021-se-2025, treated-only NDBI change (H1) — dono windows mein test hua, ek robustness-check ki tarah jo pehle se hi set-up thi. Paper ka central claim (Sections 4.2, 4.9, 5, 8 — ye sab PDF 21 mein aayenge) isi ek estimand ke around bana hai: kya kisi bhi window mein, koi statistically-detectable built-up-area increase hai bhi ya nahi?

Secondary, pre-specified robustness checks — control-group DiD (H4), buffer-radius sweep, leave-one-district-out aur randomization inference, aur Holm-Bonferroni correction — ye sab primary estimand ko stress-test karne ke liye hain, usse replace karne ke liye nahi. In mein se koi bhi study ka headline-finding redefine nahi karta, sirf isliye kyunki wo usse aseemat hai.

Secondary, independent research questions — night-lights change (RQ3, H1 jaisa hi tareeka), border-proximity correlation (H3), teen-point multi-year trend, aur budget-correlation comparison (RQ2, explicitly descriptive) — ye sab alag research-questions ka jawab dete hain, chahe jo bhi dikhaayein, null ho ya positive, wo primary NDBI estimand ke conclusion ko na mazboot karte hain na kamzor.

Secondary, ab-stress-tested finding — summer-window H3 NDBI border-proximity correlation (Section 4.5), jo sirf Section 4.2 ke data-correction ke baad hi saamne aaya. Ye tab se usi leave-one-out aur randomization-inference checks se guzar chuka hai jo primary estimand ke robustness-suite ne maange the, aur dono mein zinda bacha hai (Section 7.6, Development Log Entry 32). Phir bhi, ye H1 ke jawab mein kisi bhi tarah wapas feed nahi karta.

Preregistration Ka Substitute — Ek Imaandaar Disclosure

Ye study kisi public registry par preregister nahi hui thi data-collection se pehle, aur ye section isse chhupata nahi. Iske badle mein: ANALYSIS_FREEZE.md, exact

commit, data-extraction-state, aur tareekh (sabse recent 2026-09-17, Development Log Entries 25-35 ke through carry-forward) record karta hai jispar is paper ke headline numbers bane hain, aur upar bataayi gayi estimand-hierarchy usi freeze par tay hui thi, baad mein data ke hisaab se adjust nahi hui. Us freeze ke baad se har deviation — dono reversals jo ye paper khule taur par batata hai (Sections 4.2 aur 4.6, PDF 21 mein) shaamil — dated hai aur ek specific Development Log entry tak traceable hai, chupke se andar nahi ghusaya gaya.

Ye ek asli preregistration se kamzor guarantee hai, kyunki estimand-hierarchy khud usi vyakti ne likhi jisne analysis chalaya, na ki kisi independent registry mein pehle se file ki gayi — aur ek skeptical reader isse uske hisaab se weigh karne ka poora haq rakhta hai. Ye yahan is study ki apni history se jo best-possible substitute ban sakta tha, usi ki tarah pesh kiya gaya hai — ek equivalent ki tarah nahi.

In Sab Ko Jodkar Dekho

Is PDF mein humne poore paper ka 'setup' seekha: Abstract ne poore result ka nichod diya, Introduction ne research-questions aur hypotheses set kiye, Literature Review ne theoretical buniyad aur citation-trail di, aur Methodology ne har number ke peeche ki kitchen kholi — geocoding se lekar DAG-diagram tak, har jagah history bunte hue. Teen alag-alag imaandaar moments is PDF mein dikhe: citation-audit ki galti aur fix (2.2), summer-vs-full-year window ka reversal aur uski sequencing ka ek zaroori nuance (3.5), aur poori estimand-hierarchy ka preregistration-substitute wala khula disclosure (3.12). Har jagah paper khud apni history batane se nahi katata — aur yahi sabse zaroori seekh hai poore BORDER_OPTICS project se: acchi research wo nahi jo kabhi galat na ho, acchi research wo hai jo apni galtiyan ko khule taur par record kare, fix kare, aur dikhaaye.

RECAP — PDF 20 Mein Kya Seekha: (1) Paper ka title do hisso mein tuta hai — VVP-I ka naam, aur asli methodological star, compositing-window-robustness. (2) Abstract ke chaar paragraphs — headline null (H1/H4/buffer/leave-one-out/randomization, saath 154-village Sikkim-reversal aur night-lights-contamination-fix ki history), H3 border-proximity ka ulta-result (jo ab fully stress-tested hai), triangulation ki khuli tension (SAR sehmata, Dynamic World asehmata) aur ek doosri jo tension nahi bug nikli (QA60 vs SCL), aur budget ka descriptive-only kissa. (3) Introduction ke 5 research questions aur 4 hypotheses. (4) Literature Review — securitization theory, VVP ka bikhra hua policy-trail (2026-09-18 citation-audit correction samet), NDBI/VIIRS ki theoretical buniyad, aur compositing-window-sensitivity ko jaan-bujh kar design-feature ki tarah frame karna. (5) Poori Methodology, 3.1 se 3.12 — study-design, data-sources, geocoding, satellite-extraction, dual-window history (aur ek imaandaar nuance paper ke apne 'pre-specified' claim mein), border-proximity, DiD regression formula term-by-term, multi-year trend, buffer-radius robustness, poore statistical-testing ka recap, causal DAG (kya district-FE net-out karta hai vs kya nahi), SUTVA/spillover ka conservative-bias argument, aur estimand-hierarchy ka preregistration-substitute disclosure.

Aage, PDF 21 mein paper ka doosra hissa shuru hoga — Results, Discussion, Limitations, Future Work, aur Conclusion (Section 4 se 8 tak) — jahan ye poori Methodology jo numbers deti hai, unhe pehli baar poori tarah, ek-ek section karke dekha jaayega.